



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2016 HIGHER SCHOOL CERTIFICATE ASSESSMENT TASK # 2

Mathematics Extension 1

General Instructions

- Reading Time – 5 Minutes.
- Working time – 90 Minutes.
- Write using black or blue pen. .
- Board approved calculators maybe used.
- The Board Approved Reference Sheet is Provided.
- **ALL** necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for untidy or badly arranged work.
- Answer in simplest **EXACT** form unless otherwise instructed.
- Attempt Questions 1–9
- The mark value of each question is shown on the right hand side.
- Attempt Questions 1–6 on the Multiple Choice answer sheet provided.
- Each section is to be answered in a **NEW** writing booklet, clearly labelled **Question 7, Question 8** and **Question 9**.
- Total Marks – 63

Examiner

A. Wang

Section I – Multiple Choice

6 Marks

Attempt Questions 1–6

Use the multiple-choice answer sheet for Questions 1–6

1 What is the derivative of $y = 3\sin x - 4\cos x$?

(A) $\frac{dy}{dx} = 3\cos x - 4\sin x$

(B) $\frac{dy}{dx} = 3\cos x + 4\sin x$

(C) $\frac{dy}{dx} = -3\cos x + 4\sin x$

(D) $\frac{dy}{dx} = -3\cos x - 4\sin x$

2 In how many ways can 12 students of unique height be arranged in a line, so that the tallest and the shortest student never come together?

(A) $11! \times 10$

(B) $10! \times 10$

(C) $11! \times 9$

(D) $10! \times 11$

3 Consider the function $f(x) = x^4 + 3x^2 - x - 5$. It has one root at $x = -1$. Take $x = 2$ as a first approximation for this root.

Using two applications of Newton's method, which of the following is a better approximation for the root?

(A) $x = -1.257$

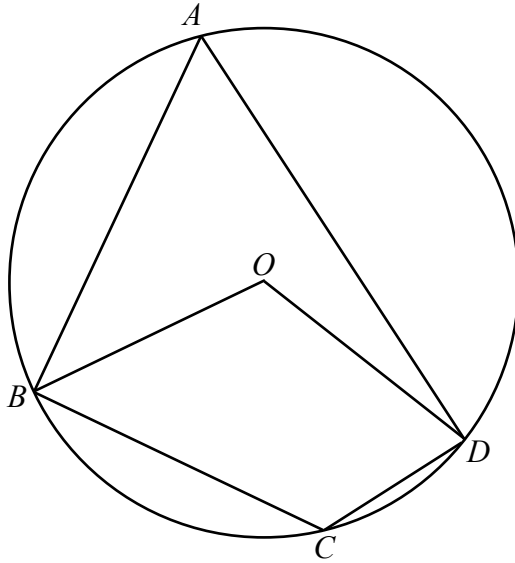
(B) $x = 1.257$

(C) $x = -1.512$

(D) $x = 1.512$

- 4 In the diagram A, B, C and D are points on a circle with centre O .

$$\angle BAD = x^\circ \text{ and } \angle BOD = \angle BCD.$$



NOT TO SCALE

What is the value of x ?

- (A) 75°
(B) 120°
(C) 90°
(D) 60°
- 5 What is the solution to $\frac{{}^nC_4}{{}^{n-2}C_2} = 1$?
- (A) 5
(B) 4
(C) 6
(D) 2
- 6 What is the length of the chord of the parabola $x^2 = 4ay$ passing through the vertex and having slope $\tan \alpha$?
- (A) $4a \operatorname{cosec} \alpha \cot \alpha$
(B) $4a \sec \alpha \tan \alpha$
(C) $4a \cos \alpha \cot \alpha$
(D) $4a \sin \alpha \tan \alpha$

Section II

57 marks

Attempt Questions 7–9

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

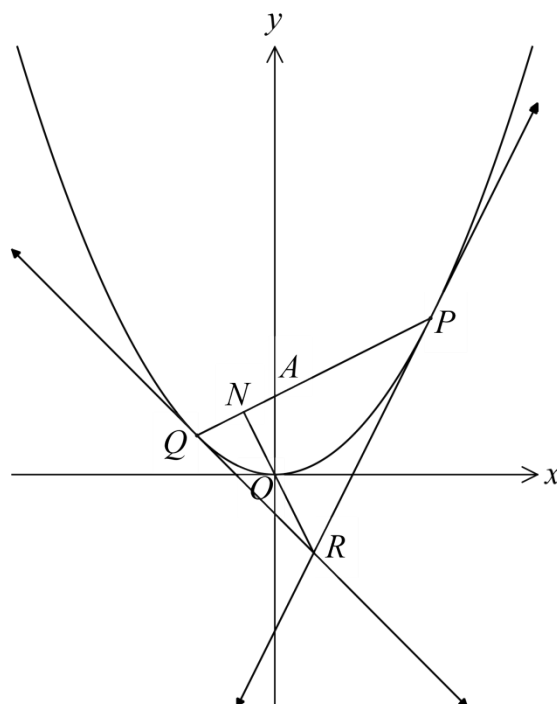
In Questions 7–9, your responses should include relevant mathematical reasoning and/or calculations.

Question 7 (19 marks)

Start a NEW Writing Booklet

- (a) The diagram below shows a parabola defined by the parametric equations

$$x = 2t \text{ and } y = t^2$$

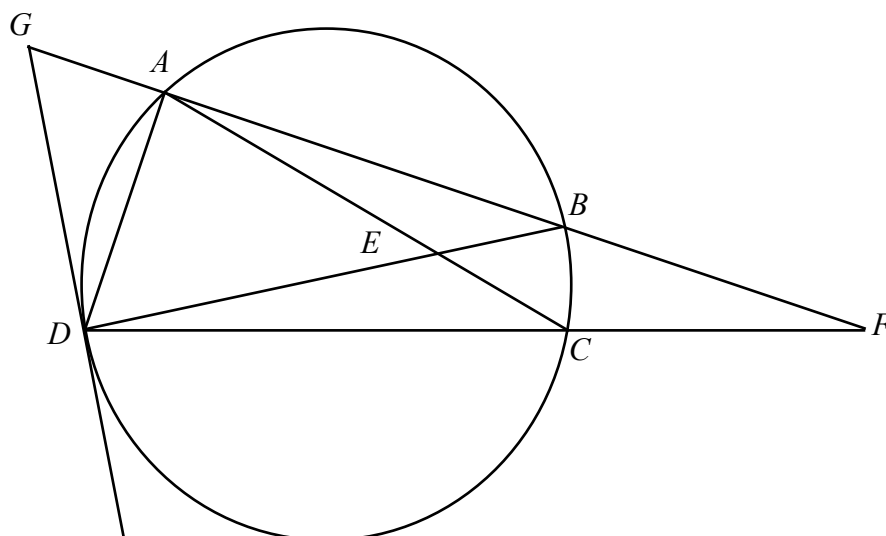


- i) Write down the equations of the tangents at the points $P(2p, p^2)$ and $Q(2q, q^2)$. 1
- ii) Show that the point of intersection of the two tangents is $R(p + q, pq)$. 2
- iii) Show that the equation of the chord PQ is $(p + q)x - 2y - 2pq = 0$. 2
- iv) Points P and Q move on the parabola in such a way that $pq = -2$. 1
Prove that the chord PQ always passes through the point $A(0, 2)$.
- v) N is the intersection of the chord PQ and the line through R and O . 2
Show that RN is perpendicular to PQ .

Question 7 continues on the next page

Question 7 (continued)

- b) In the diagram below, DG is a tangent to the circle at D .
 $GABF$ and DCF are straight lines.

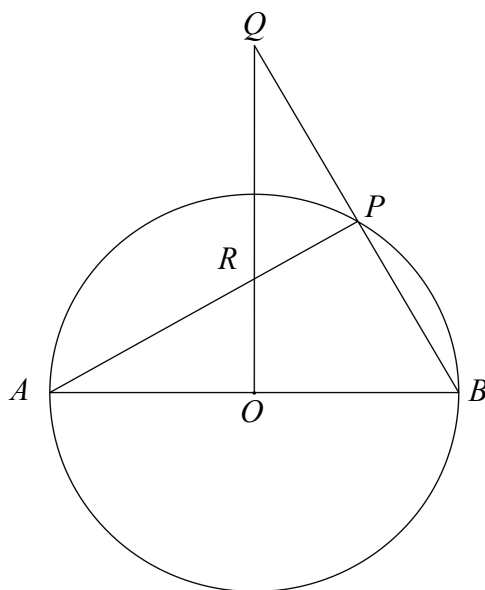


- i) Copy the diagram into your writing booklet.
- ii) Prove $2 \times \angle ADG = \angle BEC + \angle BFC$. 3
- c) Evaluate $\lim_{x \rightarrow 0} \frac{1}{x} \sin \frac{x}{2\pi}$. 1
- d) Find:
- i) $\frac{d}{dx} \cos(\sqrt{x})$ 2
- ii) $\frac{d}{dx} \tan(\sin 3x)$ 2
- e) Show that $\frac{d}{dx} (\sec x) = \sec x \tan x$ 1
- f) Find the area bounded by the curve $y = \sin\left(\frac{x}{2}\right)$, the lines $x = -\pi$, $x = \pi$, and the x -axis. 2

End of Question 7

Question 8 (19 marks)**Start a NEW Writing Booklet**

- a) O is the centre of the circle APB .
 BPQ , ORQ , ARP and AOB are straight lines. $\angle QOB = 90^\circ$.



- i) Copy the diagram into your writing booklet.
- ii) Prove that A , O , P and Q are concyclic points. **3**
- b) In how many rearrangements of the letters of the word SCINTILLATING will no two 'T's appear together? **2**
- c) The function $f(x) = x^2 - \ln(x + 1)$ has one root between 0.5 and 1.
- i) Show that the root lies between 0.7 and 0.8. **2**
- ii) Hence use the halving interval method to find the value of the root correct to 1 decimal place. **2**

Question 8 continues on the next page.

Question 8 (continued)

- d) Let A be a point on the parabola $x^2 = 4by$ whose coordinates are $(2bt, bt^2)$.
Tangents are drawn from A to another parabola $x^2 = 4ay$, where $b > a$.
These tangents touch the parabola $x^2 = 4ay$ at $P(2ap, ap^2)$ and $Q(2aq, aq^2)$.

From P and Q , normals are drawn to intersect at N .

- i) Sketch a diagram to represent the above information. 1
- ii) Show that $bt^2 - 2bpt + ap^2 = 0$ and $bt^2 - 2bqt + aq^2 = 0$ 2
- iii) Show that the coordinates of N are given by
 $x = -apq(p + q), y = a[(p + q)^2 - pq + 2]$. 2
- iv) From part ii) above, show that $a(p + q) = 2bt$ and $apq = bt^2$. 2
- v) Hence show that the locus of N , the intersection of the normals to $x^2 = 4ay$, 3
is the curve

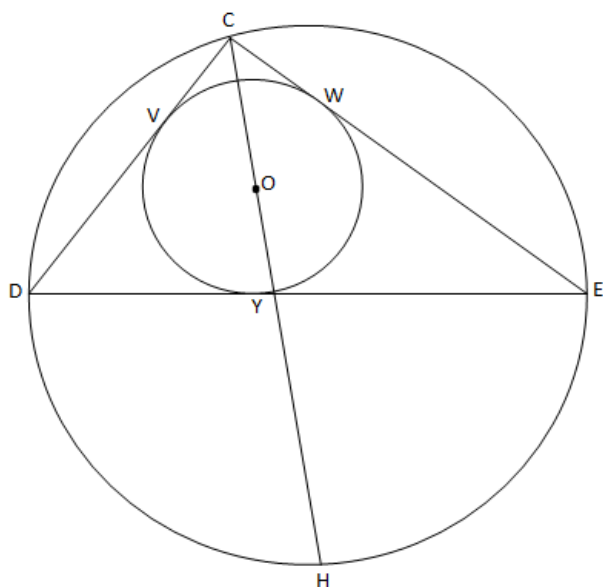
$$x^2(4b - a)^3 = 4ab(y - 2a)^3.$$

End of Question 8

Question 9 (19 marks)

Start a NEW Writing Booklet

- a) The incircle of triangle CDE has centre O and touches the sides of $\triangle CDE$ at V , W and Y . The circumcircle of triangle CDE meets CO produced to H .



- i) Copy the diagram to your writing booklet.
 - ii) Prove that H is the midpoint of arc DE . 2
 - iii) Prove that $\triangle ODV \equiv \triangle ODY$. 2
 - iv) Prove that $\angle ODH = \angle DOH$. 2
 - v) Prove $HD = HO$. 1
- b)
- i) Simplify $\cos(A - B) - \cos(A + B)$ 1
 - ii) Prove by the method of mathematical induction that 3

$$\sin w + \sin 3w + \sin 5w + \dots + \sin(2n - 1)w = \frac{1 - \cos 2nw}{2 \sin w}$$
for a constant w and for integers $n \geq 1$.
- c) The straight line $y = mx + b$ meets the parabola $x^2 = 4y$ at two points L and N . M is the midpoint of LN .
- i) Find the coordinates of M in terms of m and b . 3
 - ii) Find the locus of M if $b = -2$. 3
 - iii) What are the restrictions on the domain of M ? Justify your answer. 2

End of paper



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2016

HSC Task #2

Mathematics Extension 1

Suggested Solutions & Markers' Comments

QUESTION	Marker
1 – 6	–
7	EC
8	BK
9	AF

Multiple Choice Answers

1. B
2. A

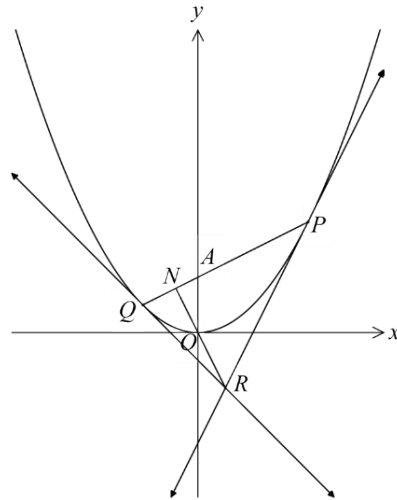
3. B
4. D

5. B
6. B

Question 7

Solutions

- (a) The diagram below shows a parabola defined by the parametric equations
 $x = 2t$ and $y = t^2$



- i) Write down the equations of the tangents at the points $P(2p, p^2)$ and $Q(2q, q^2)$. 1

Tangent at P : $y = px - p^2$

Tangent at Q : $y = qx - q^2$

- ii) Show that the point of intersection of the two tangents is $R(p + q, pq)$. 2

$$y = px - p^2 \quad \text{--- (1)}$$

$$y = qx - q^2 \quad \text{--- (2)}$$

Equating (1) and (2): $(p - q)x = p^2 - q^2$

$$\therefore (p - q)x = (p - q)(p + q)$$

$$\therefore x = p + q$$

Substitute into (1): $y = p(p + q) - p^2$

$$\therefore y = pq$$

$\therefore$ the point of intersection is $(p + q, pq)$.

- iii) Show that the equation of the chord PQ is $(p + q)x - 2y - 2pq = 0$. 2

$$\begin{aligned} m_{PQ} &= \frac{q^2 - p^2}{2q - 2p} \\ &= \frac{(q - p)(q + p)}{2(q - p)} \\ &= \frac{1}{2}(p + q) \end{aligned}$$

$$\therefore y - q^2 = \frac{1}{2}(p + q)(x - 2q)$$

$$\therefore 2y - 2q^2 = (p + q)x - 2q(p + q)$$

$$\therefore (p + q)x - 2y + 2q^2 - 2q(p + q) = 0$$

$$\therefore (p + q)x - 2y - 2pq = 0$$

- (a) iv) Points P and Q move on the parabola in such a way that $pq = -2$. 1
 Prove that the chord PQ always passes through the point $A(0, 2)$.

Substitute $pq = -2$ into $(p + q)x - 2y - 2pq = 0$

$$\therefore (p + q)x - 2y + 4 = 0$$

y -intercept when $x = 0$

$$\therefore -2y + 4 = 0$$

$$\therefore y = 2$$

i.e. $(0, 2)$ always lies on the chord PQ .

- v) N is the intersection of the chord PQ and the line through R and O . 2
 Show that RN is perpendicular to PQ .

$$m_{RN} = \frac{pq}{p + q}$$

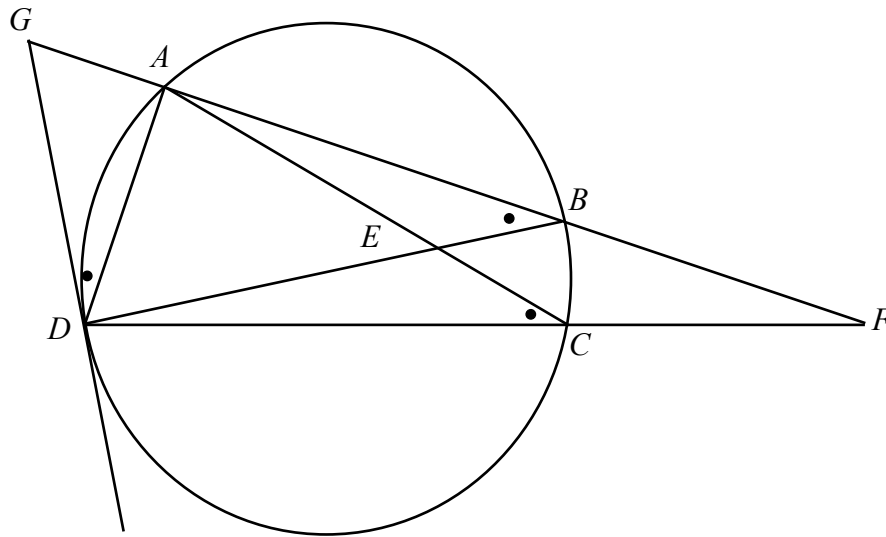
$$= \frac{-2}{p + q}$$

$$m_{PQ} \times m_{RN} = \frac{1}{2}(p + q) \times \frac{-2}{p + q}$$

$$= -1$$

$$\therefore PQ \perp RN$$

- b) In the diagram below, DG is a tangent to the circle at D .
 $GABF$ and DCF are straight lines.



- i) Copy the diagram into your writing booklet.

- ii) Prove $2 \times \angle ADG = \angle BEC + \angle BFC$.

3

$$\angle ADG = \angle ABE \quad (\text{angle between tangent and chord})$$

$$\text{Similarly, } \angle ADG = \angle ACD$$

$$\angle BEC + \angle BFC = 2\pi - (\angle EBF + \angle ECF) \quad (\text{angle sum of quad. } BECF)$$

$$\angle EBF = \pi - \angle ABE \quad (\text{straight } \angle)$$

$$\text{Similarly, } \angle ECF = \pi - \angle ACD$$

$$\therefore \angle BEC + \angle BFC = \angle ACD + \angle ABE$$

$$\therefore \angle BEC + \angle BFC = 2 \times \angle ADG$$

- c) Evaluate $\lim_{x \rightarrow 0} \frac{1}{x} \sin \frac{x}{2\pi}$.

1

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1}{x} \sin \frac{x}{2\pi} &= \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2\pi}}{\frac{x}{2\pi}} \\ &= \lim_{x \rightarrow 0} \left(\frac{\sin \frac{x}{2\pi}}{\frac{x}{2\pi}} \times \frac{1}{2\pi} \right) \\ &= \frac{1}{2\pi} \times \lim_{x \rightarrow 0} \left(\frac{\sin \frac{x}{2\pi}}{\frac{x}{2\pi}} \right) \\ &= \frac{1}{2\pi} \end{aligned}$$

d) Find:

i) $\frac{d}{dx} \cos(\sqrt{x})$ 2

$$\begin{aligned} \frac{d}{dx} \cos(\sqrt{x}) &= \frac{d}{dx} \cos(x^{\frac{1}{2}}) \\ &= -\sin(x^{\frac{1}{2}}) \times \frac{1}{2} x^{-\frac{1}{2}} \\ &= -\frac{\sin(x^{\frac{1}{2}})}{2x^{\frac{1}{2}}} \end{aligned}$$

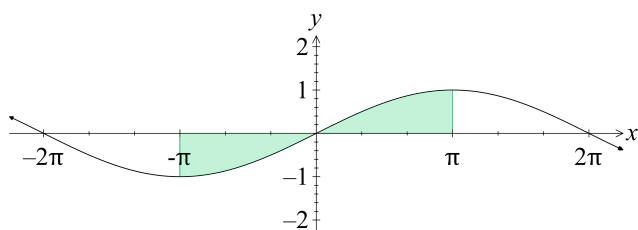
ii) $\frac{d}{dx} \tan(\sin 3x)$ 2

$$\begin{aligned} \frac{d}{dx} \tan(\sin 3x) &= \sec^2(\sin 3x) \times \frac{d}{dx}(\sin 3x) \\ &= 3 \cos 3x \sec^2(\sin 3x) \end{aligned}$$

e) Show that $\frac{d}{dx}(\sec x) = \sec x \tan x$ 1

$$\begin{aligned} \frac{d}{dx}(\sec x) &= \frac{d}{dx}(\cos x)^{-1} \\ &= -(\cos x)^{-2} \times (-\sin x) \\ &= \frac{\sin x}{\cos^2 x} \\ &= \frac{\sin x}{\cos x} \times \frac{1}{\cos x} \\ &= \tan x \sec x \end{aligned}$$

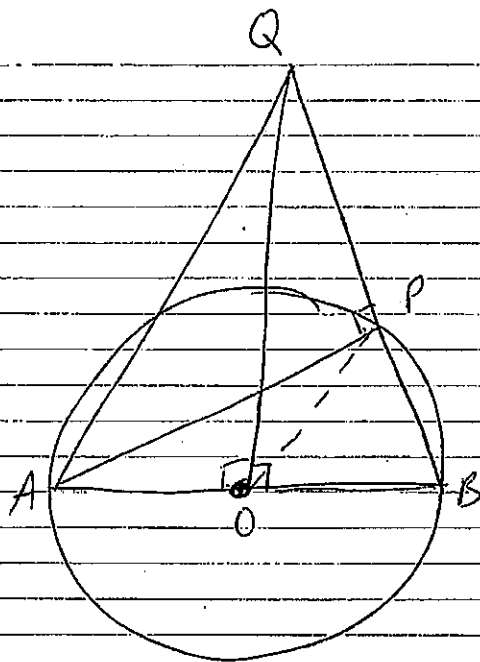
f) Find the area bounded by the curve $y = \sin\left(\frac{x}{2}\right)$, the lines $x = -\pi$, $x = \pi$, and the x -axis. 2



$$\begin{aligned} \text{Area} &= 2 \times \int_0^{\pi} \sin\left(\frac{x}{2}\right) dx \\ &= 2 \times \left[-2 \cos\left(\frac{x}{2}\right) \right]_0^{\pi} \\ &= -4 \left[\cos\left(\frac{\pi}{2}\right) - \cos 0 \right] \\ &= 4 \end{aligned}$$

Q8.

(a) (i)



$\angle AOB = 90^\circ$
(given)

(ii) Prove A, O, P, Q are concyclic.

$\angle APB = 90^\circ$ (Angle in a semi-circle) ✓

$\Rightarrow \angle APQ = 90^\circ$ (Supplementary $\angle$ s on straight line)
and $\angle AOQ = 90^\circ$ (same reason)

Then $\angle APQ = \angle AOQ = 90^\circ$ ✓

Then P and O are angles at the circumference of a circle with QA as the diameter i.e. they are angles in a semi-circle ✓

$\therefore AOPQ$ is a circle and A, O, P, Q are concyclic. ✓

Mostly done well. Some did a longer solution using congruent triangles.

(3)

8(b) SCINTILLATING

IS
IC
3I
2N
2T
2L
1A
1G
13 letters

Put all 10 letters, apart from the I's, down first

$= \frac{10!}{2!2!2!}$ ways. ✓

|_|_|_|_|_|_|_|_|_|

Then there are 11 possible spaces in which to place the 3 I's $\Rightarrow$ choose 3 spaces from 11.

$= {}^{11}C_3 \text{ ways} = \frac{11!}{3!8!}$ ways.

Then total ways $= \frac{10!}{2!2!2!} \times \frac{11!}{3!8!}$ ✓ (2)
 $= 74844000$ ways.

About two-thirds of the students used the spaces method and got the correct answer.

Others tried to do cases but none did so successfully.

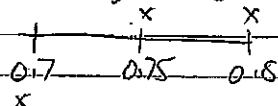
8(c)(i) $y = x^2 - \ln(x+1)$ Most did this question well.

$$f(0.7) = -0.0406 < 0$$

$$f(0.8) = 0.0522 > 0$$

$\therefore$ there is a value between 0.7 and 0.8 for which $f(x) = 0$ since $f(x)$ is a continuous function

$$(ii) f\left(\frac{0.7+0.8}{2}\right) = f(0.75) = 0.00288 > 0$$



$\therefore$ root is closer to $x = 0.7$ (to 1 dp).

Many students spent time doing multiple applications of the halving the interval method before realising that they only needed the x-value to 1 dp. A few used the wrong method (ie Newton's Method)

8(d)(i)

A majority of students were unable to draw the correct diagram and then could not achieve full marks in (ii)

(ii) Show $bt^2 - 2bpt + ap^2 = 0$ and $bt^2 - 2bqt + aq^2 = 0$

$$y = \frac{x^2}{4a}$$

$$y' = \frac{x}{2a}$$

$$\text{At } P, y' = \frac{2ap}{2a} = p$$

$$\therefore \text{eqn of tangent at } P \text{ is } y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2$$

But A lies on tangent ie $(2bt, bt^2)$ satisfies eqn

$$\Rightarrow bt^2 = 2btp - ap^2$$

$$\Rightarrow bt^2 - 2btp + ap^2 = 0$$

Similarly, tangent at Q is $y = qx - aq^2$ and A lies on tangent

$$\Rightarrow bt^2 - 2btq + aq^2 = 0$$

All those who did not have correct diagram in (i) received 1 mark for the correct process of finding a tangent at A and the equations through the other points.

8(d)

(iii) Equation of Normal at P:

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$yp - ap^3 = -x + 2ap$$

$$yp + x = ap^3 + 2ap \quad (1)$$

Similarly the Normal at Q has equation

$$yq + x = aq^3 + 2aq \quad (2)$$

$$(1) - (2) \Rightarrow y(p - q) = a(p^3 - q^3) + 2a(p - q)$$

$$y(p - q) = a(p - q)(p^2 + pq + q^2) + 2a(p - q)$$

$$\Rightarrow y = a(p^2 + pq + q^2) + 2a \quad \text{since } p \neq q$$

$$\Rightarrow y = a[(p + q)^2 - pq + 2] \quad \# \checkmark$$

$$\text{Also } q \times (1) \Rightarrow ypq + qx = aqp^3 + 2apq \quad (3)$$

$$\text{and } p \times (2) \Rightarrow ypq + px = apq^3 + 2apq \quad (4)$$

$$(4) - (3) \Rightarrow (p - q)x = apq(q^2 - p^2)$$

$$(p - q)x = -apq(p^2 - q^2)$$

$$x = -apq(p + q) \quad \# \checkmark$$

Done well
overall.

(2)

8(d)

$$(iv) \quad bt^2 - 2bpt + ap^2 = 0 \quad (1)$$

$$bt^2 - 2bqt + aq^2 = 0 \quad (2)$$

$$p \times (2) \Rightarrow bpt^2 - 2bpqt + apq^2 = 0 \quad (3)$$

$$q \times (1) \Rightarrow bq^2t^2 - 2bpqt + ap^2q = 0 \quad (4)$$

$$(3) - (4) \Rightarrow bt^2(p - q) + apq(q - p) = 0$$

$$bt^2(p - q) = apq(p - q)$$

$$\Rightarrow bt^2 = apq \quad (p \neq q)$$

$$pq = \frac{bt^2}{a} \quad (5) \quad \checkmark \quad (2)$$

$$\text{Also } (1) - (2) \Rightarrow -2bpt + 2bqt + a(p^2 - q^2) = 0$$

$$-2bt(p - q) = -a(p^2 - q^2)$$

$$2bt = a(p + q) \quad \checkmark$$

Most students did this
question successfully.

$$\frac{2bt}{a} = p + q \quad (6) \quad \#$$

$$(v) \quad N = (-apq(p + q), a[(p + q)^2 - pq + 2]) \quad (8)$$

$$\Rightarrow x = -apq(p + q) \quad (7) \text{ and } y = a[(p + q)^2 - pq + 2]$$

sub (5)(6) into (7) from part (iv)

$$\Rightarrow x = -a\left(\frac{bt^2}{a}\right)\left(\frac{2bt}{a}\right)$$

$$\Rightarrow x = -\frac{2b^2t^3}{a}$$

$$\Rightarrow t^3 = -\frac{ax}{2b^2}$$

$$\Rightarrow t = \sqrt[3]{-\frac{ax}{2b^2}} \quad (9) \quad \checkmark$$

8. (d) (v) (continued)

Sub (5), (6) in (8)

$$y = a \left[\frac{4b^2 t^2}{a^2} - \frac{bt^2}{a} + 2 \right]$$

$$y = \frac{4b^2 t^2}{a} - bt^2 + 2a$$

$$y - 2a = t^2 \left(\frac{4b^2}{a} - b \right)$$

Sub in $t = \frac{a^{\frac{1}{3}} x^{\frac{1}{3}}}{2^{\frac{1}{3}} b^{\frac{2}{3}}}$

$$\Rightarrow y - 2a = \frac{a^{\frac{2}{3}} x^{\frac{2}{3}}}{2^{\frac{2}{3}} b^{\frac{4}{3}}} \left(\frac{4b^2}{a} - b \right) \checkmark$$

Cube both sides:

$$(y - 2a)^3 = \frac{a^2 x^2}{4b^4} \left(\frac{4b^2}{a} - b \right)^3$$

$$(y - 2a)^3 = \frac{a^2 x^2}{4b^4} \left[\frac{b(4b - a)}{a} \right]^3$$

$$(y - 2a)^3 = \frac{a^2 x^2}{4b^4} \frac{b^3}{a^3} (4b - a)^3$$

$$(y - 2a)^3 = \frac{x^2}{4ab} (4b - a)^3$$

$$4ab(y - 2a)^3 = x^2(4b - a)^3 \checkmark \text{ (3) \#}$$

8(d)(v) (continued)

Alternate Method

$$\text{From } t^3 = \frac{-ax}{2b^2} \Rightarrow t^6 = \frac{x^2 a^2}{4b^4} \quad (1)$$

$$\text{Also from } y - 2a = t^2 \left(\frac{4b^2}{a} - b \right)$$

$$y - 2a = t^2 \left(\frac{4b^2 - ab}{a} \right)$$

$$\Rightarrow t^2 = \frac{(y - 2a)a}{4b^2 - ab}$$

$$t^2 = \frac{(y - 2a)a}{b(4b - a)}$$

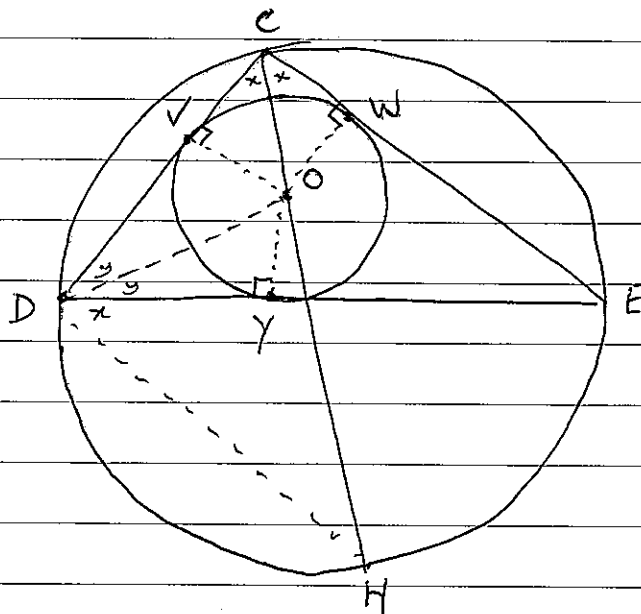
$$\Rightarrow t^6 = \frac{(y - 2a)^3 a^3}{b^3 (4b - a)^3} \quad (2)$$

$$\text{Then } (1) = (2) \Rightarrow \frac{x^2 a^2}{4b^4} = \frac{(y - 2a)^3 a^3}{b^3 (4b - a)^3}$$

$$\Rightarrow x^2 (4b - a)^3 = 4ab (y - 2a)^3 \#$$

Only a few students got this question out correctly. 1 mark was assigned if students correctly expressed x and y in terms of a, b and t

a) i)



ii) In Δ 's cov & cow

$OV = OW$ (equal radii)

$$\hat{CVO} = \hat{CWO} = 90^\circ \quad (\text{radius} \perp \text{tangent})$$
$$\Delta C_{OV} \equiv \Delta C_{OW} \text{ (RHS)}$$
$$\hat{v}_{CO} = \hat{w}_{CO} \text{ (corresponding angles, } \Delta COV \equiv \Delta COW)$$

$\text{arc DH} = \text{arc HE}$ (arcs that subtend equal angles are equal)

$\therefore H$ is the midpoint of arc DE .

iii) In Δ 's $DVO \neq DYO$

$Or = Oy$ (equal radii)

$$\hat{D}\hat{V}O = \hat{D}\hat{Y}O = 90^\circ \text{ (radius } \perp \text{ tangent)}$$
$$\Delta DVO \equiv \Delta DYO \quad (\text{RHS})$$

iv) let $\hat{VCO} = \hat{WCO} = x$ (proven in (ii))

let $\hat{VDO} = \hat{YDO} = y$ (corresponding angles, $\triangle PVO \cong \triangle DVO$)

$$\hat{HDE} = x = \hat{HCE} \quad (\text{angles in the same segment})$$
$$\angle DOH = x + y \text{ (exterior angle of } \triangle DOC)$$
$$\hat{HDO} = x + y$$

$$\therefore \hat{ODH} = \hat{DOH}$$

v) $HD = HO$ (sides opposite equal angles, $\triangle ODH$)

COMMENTS:

- Part (v) should have been an easy mark for all students regardless of their success with the other questions.
- Part (iii) could be proven using the congruency tests SSS, SAS, RHS and was generally well attempted.
- Very few students made any progress with part (iv).
- There were students who oversimplified the diagram by making Y the point where CO meets DE .

$$\begin{aligned} \text{b) i) } & \cos(A-B) - \cos(A+B) \\ &= \cancel{\cos A \cos B} + \sin A \sin B - (\cancel{\cos A \cos B} - \sin A \sin B) \\ &= 2 \sin A \sin B \end{aligned}$$

ii) Prove true for $n=1$

$$\text{LHS} = \sin w$$

$$\text{RHS} = \frac{1 - \cos 2w}{2 \sin w}$$

$$= \frac{1 - (1 - 2 \sin^2 w)}{2 \sin w}$$

$$= \sin w$$

$$\text{LHS} = \text{RHS}$$

$\therefore$ true for $n=1$

Assume true for $n=k$ where $k \in \mathbb{N}$

$$\sin w + \sin 3w + \dots + \sin(2k-1)w = \frac{1 - \cos 2kw}{2 \sin w}$$

Prove true for $n=k+1$

$$\text{ie } \sin w + \sin 3w + \dots + \sin (2k-1)w + \sin (2k+1)w = \frac{1 - \cos 2(k+1)w}{2 \sin w}$$

$$\text{LHS} = \sin w + \sin 3w + \dots + \sin (2k-1)w + \sin (2k+1)w$$

$$= \frac{1 - \cos 2kw}{2 \sin w} + \sin (2k+1)w$$

$$= \frac{1 - \cos 2kw + 2 \sin (2k+1)w \sin w}{2 \sin w}$$

$$= \frac{1 - \cancel{\cos 2kw} + \cancel{\cos 2kw} - \cos (2k+2)w}{2 \sin w}$$

$$= \frac{1 - \cos 2(k+1)w}{2 \sin w}$$

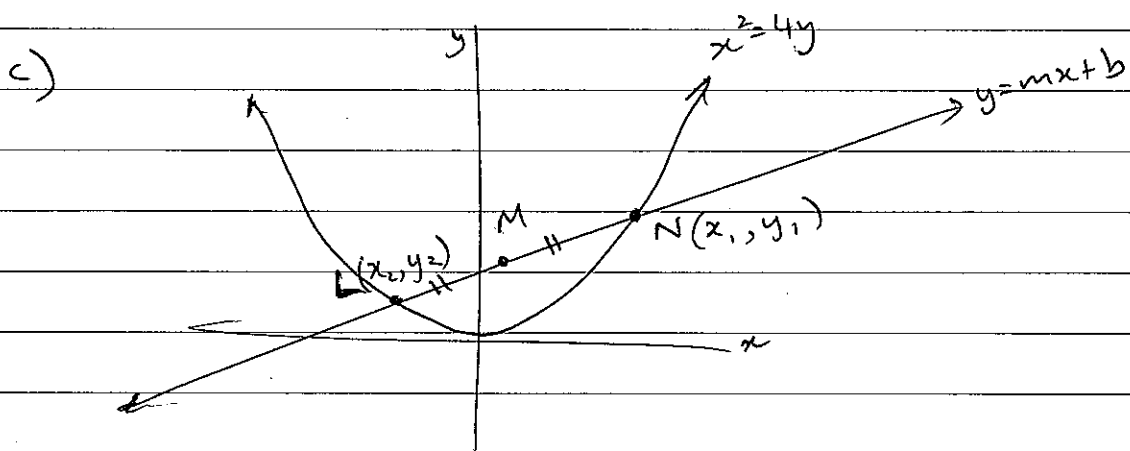
$$= \text{RHS}$$

$\therefore$ true for $n=k+1$

$\therefore$ true by induction for integers $n \geq 1$.

COMMENT:

- Students should look to use part (i)
- Students should not assume what they are required to prove.



$$i) \quad y = mx + b \quad \text{--- (1)}$$

$$y = \frac{x^2}{4} \quad \text{--- (2)}$$

sub ① into ②

$$\frac{x^2}{4} = mx + b$$

$x^2 - 4mx - 4b = 0$ has roots x_1 & x_2 .

$$x_1 + x_2 = -\frac{B}{A}$$

$$= -\frac{(-4m)}{1}$$

$$= 4m$$

$$x \text{ coordinate of } M \text{ is } \frac{x_1 + x_2}{2} = \frac{4m}{2}$$

$$= 2m$$

M lies on $y = mx + b$

$$y \text{ coordinate of } M \text{ is } y = m(2m) + b$$

$$y = 2m^2 + b$$

$\therefore M$ has coordinates $(2m, 2m^2 + b)$

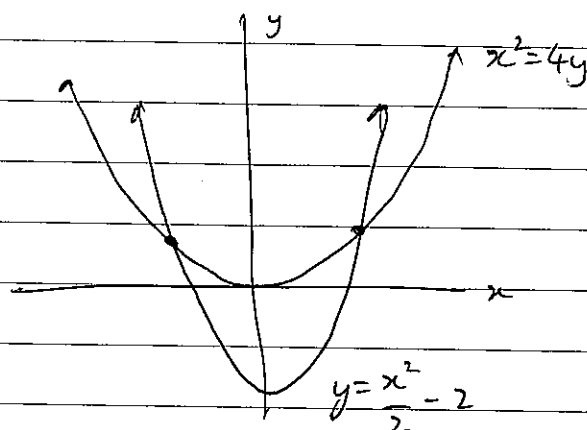
$$ii) \quad x = 2m$$

$$y = 2m^2 + (-2) \quad \text{when } b = -2$$

$$y = 2\left(\frac{x}{2}\right)^2 - 2$$

$$y = \frac{x^2}{2} - 2 \quad \text{which is a parabola.}$$

(iii)



M is the midpoint of a chord & so must lie within the parabola $x^2 = 4y$.

$$y = \frac{x^2}{4} \quad \text{--- (1)}$$

$$y = \frac{x^2}{2} - 2 \quad \text{--- (2)}$$

sub (1) into (2)

$$\frac{x^2}{4} = \frac{x^2}{2} - 2$$

$$-\frac{x^2}{4} = -2$$

$$x^2 = 8$$

$$x = \pm \sqrt{8}$$

$$x = \pm 2\sqrt{2}$$

$$\therefore x > 2\sqrt{2}, x < -2\sqrt{2}$$

COMMENTS:

- Students who first found the coordinates of L & N tended to get bogged down in the algebra.

- The discriminant could have been used for part (iii).

However, it would take a little longer, since we were asked for the restriction on the domain of M not m.